



# A Peak Shaving Approach in Manufacturing Combining Machine Learning and Job Shop Scheduling

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**Abstract.** Computerized Numerical Control (CNC) plays an important role in highly autonomous manufacturing systems with multiple machine tools. The necessary Numerical Control (NC) programs to manufacture the parts are mostly written in standardized G-code. An a priori evaluation of the energy demand of CNC-based machine processes opens up the possibility of scheduling multiple jobs according to balanced energy consumption over a production period. Due to this, we present a combined Machine Learning (ML) and Job-Shop-Scheduling (JSS) approach to evaluate G-code for a CNC-milling process with respect to the energy demand of each G-command. The ML model training data are derived by the Latin hypercube sampling (LHS) method facing the main G-code operations G00, G01, and G02. The resulting energy demand for each job enhances a JSS algorithm to smooth the energy demand for multiple jobs, as peak power consumption needs to be avoided due to its expense.

**Keywords:** CNC · Machining · Energy consumption · Machine learning · Design of experiments

## 1 Introduction

Computer Numerical Control (CNC) is a form of automation in which the mechanical actions of a machine tool or other equipment are controlled by a program containing a list of standardized instructions. CNC has become an important technology in construction, automotive, aerospace, and medical industries, both for part manufacturing and product design [1].

The energy crisis in Europe and, as a consequence, the increasing energy cost force manufacturing companies to analyze their processes in terms of energy efficiency. In Germany, the manufacturing industry causes around 24 % of nationwide  $CO_2$  emissions. According to Anderberg and Kara [2], the energy cost of

CNC machining accounts for 1–6% of the total machining expense, excluding the escalation rate of energy costs. Energy consumption in a CNC machine tool originates from the spindle operation, axis feed, servos system, tool change system, and cooling equipment. Traditionally, in a company that deploys a large fleet of CNC machines and in production planning, customer orders are scheduled by optimizing their makespan, lead times, or downtime of resources. Scheduling a series of orders over a defined production period (e.g. one day) can lead to electricity load peaks and can, therefore, cause higher costs. This is because, in addition to the costs for electricity consumption, there is also a considerable performance price, which the respective energy supplier calculates from the measured load peaks. Therefore, energy consumption in Job-Shop-Scheduling (JSS) has gotten a lot of attention in recent times and is a promising way of adapting to volatile electricity prices [3]. Hence, in this contribution, we present an ensemble Machine Learning (ML) model for the standard G-code operations G00 (rapid movement), G01 (linear movement), and G02 (circular movement), that predicts the energy demand for an entire Numerical Control (NC) program. The coordinates X, Y, and Z, which result in the distance traveled,  $f$  (feed rate), and  $s$  (spindle speed) are selected as input features. Additionally, for the G02 movement, the radius  $R$  is added. The aim is to estimate the energy demand of a series of jobs that should be evenly distributed over a defined production time to avoid load peaks.

In the scientific literature there are some approaches to utilizing ML to predict the energy consumption of machine tools by G-code evaluation: Borgia et al. [4] developed a Neural Network model to predict energy consumption given axis velocity, axis acceleration, spindle speed, spindle torque, and other machining parameters. They use Latin hypercube sampling (LHS) and Sobol sampling to generate data for training a two-layer Neural Network model. Bhinge et al. [5] proposed a Gaussian-Process regression model for energy prediction with three quantitative inputs (feed rate, spindle speed, and depth of cut) and two categorical inputs (cutting direction and cutting strategy). They report Normalized Mean Absolute Errors between 9.5% and 13.5% for estimating the energy for all the blocks in a given NC-code. A different approach was taken by [6]. The authors implemented a component-based energy-modeling methodology to ensure online optimization and real-time control. These energy models are able to forecast energy consumption up to the tool-path level. The effectiveness of the proposed methodology was proven on a milling machine example. Contrary to ML approaches, researchers have also provided analytical concepts to estimate the energy demand of numerical control machining. One such approach was explored by [7]. Not relying on G-code commands directly but rather on motions like auxiliary, air-cutting, or removal motions, the authors were able to estimate the power consumption of these movements. By carrying out a Design of Experiment, different lathes and CNC machine tools were tested. In the end, the authors were able to distinguish the power consumption between machines using the proposed motions. Edem et al. [8] investigated analytical models for predicting the energy demand of toolpaths. Their models aimed at estimating

the energy demand for the commands G00, G01, G02, and G03. They conclude that shorter linear path lengths with G01 and circular path segments with G02 and G03 were highly energy intensive. Another analytical method was proposed by [9]. The authors first established a correlation between numerical control codes and energy consumption components of machine tools. Subsequently, the total energy consumption of an NC machine was estimated by accumulating the energy of all machine components, like spindle, axis feed, and others. To test the prediction of their analytical model, the authors compared its results to the actual energy measurements of two workpieces. The results were very conclusive and can help to make process planning more energy efficient.

In our ML approach, we minimize the number of experiments through LHS and then train models to be ensembled to estimate the energy consumption of NC programs. As a novelty, we integrate the findings through ML into a JSS algorithm to smooth the energy required for a series of jobs planned over a production period in a factory. Due to this, the contribution of this paper can be summarized as follows:

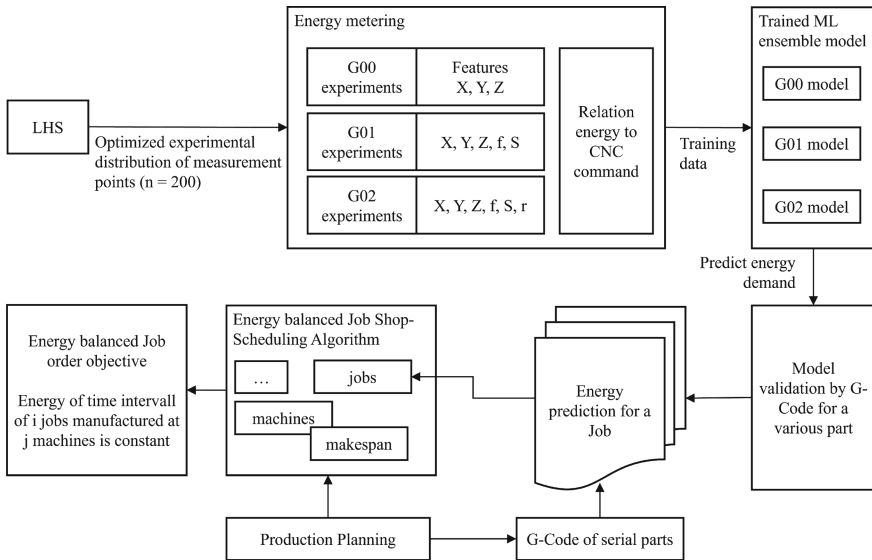
- Pioneer work to link ML and JSS to smooth energy consumption of a fleet of NC-machines during the production planning phase
- Provision of a tool to utilize order data and NC-data to optimize the production schedule
- Results of various production order scenarios indicate that our JSS algorithm is able to schedule jobs with two objectives—minimize makespan and variance of energy consumption of all working machines in a fixed time interval (e.g. 2 shift system) to reduce cost during the production planning phase by performing peak shaving as a form of load management.

The paper is structured as follows: Sect. 2 describes the research methodology and experimental setup. This includes the Design of Experiment and the generation of training data for the ML models. Based on that, Sect. 3 gives an overview of the used models and the validation results. After the ML models are established, Sect. 4 demonstrates the results these models have on the energy balancing aspect of production orders in JSS. Section 5 concludes this paper by discussing limitations and further research.

## 2 Research Methodology and Experimental Setup

Figure 1 shows the chosen research approach for energy assessment from G-code and JSS integration. First, a minimum of experimental sample points for ML model training are obtained by means of an optimal Latin Hypercube Sampling (LHS) design. LHS is a Design of Experiments method that produces a sampling plan in which points are spread across a bounded input space. A space-filling method such as LHS is useful when one of the objectives is to carry out as few experiments as possible. LHS divides each dimension of the input space into equally-sized bins and randomly generates a sample in each bin. Based on the work of Morris and Mitchel [10], we use Simulated Annealing for sample

generation [11]. Then the experiments to obtain training data are conducted for the base G-code commands G00, G01, and G02 by varying the features  $X$ ,  $Y$ , and  $Z$  for the motion path, which results in the distance as a feature. Additionally,  $f$  for the feed rate,  $S$  for spindle speed, and  $R$  for the radius of the circular interpolation of G2 were chosen as input parameters. The predictor is the energy consumption  $E$  for each G-code line. When the ensemble model is validated, the proposed method parses the G-code, calculates the energy consumption for each G-code block, and then sums up the total energy consumption for the whole NC program. For each command, the model predicts the electric energy and provides the data for each job in the JSS algorithm shown in Sect. 4.



**Fig. 1.** Principle scheme of the combined approach of ML and JSS to conduct electricity peak load shaving in manufacturing utilizing production data (NC programs with G-code and order data)

The experiments are carried out with a Stepcraft M1000 CNC milling machine. It is a three-axis machine equipped with a spindle motor. The maximum travel distance for the  $X$ -axis is 678 mm, 1043 mm in the direction of the  $Y$ -axis, and 30 mm on the  $Z$ -axis. The distance in  $Z$  is kept shorter to account for the mounted tool. The maximum feed rate  $f$  of the machine, used with the command G00, is 5100 mm/min or 85 mm/s. For commands G01 and G02 the feed rate was set between 200 and 2000 mm/min. These feed rates are enabled by the NEMA 23 stepper motor. The spindle speed is capped out at 25000 rpm and has to run at a minimum of 10000 rpm. The machine is controlled via the UCCNC machine control software. The experimental search space is shown in Table 1. The experiments are conducted without machining a material, as we tend to

**Table 1.** Machining limits defining the experimental search space

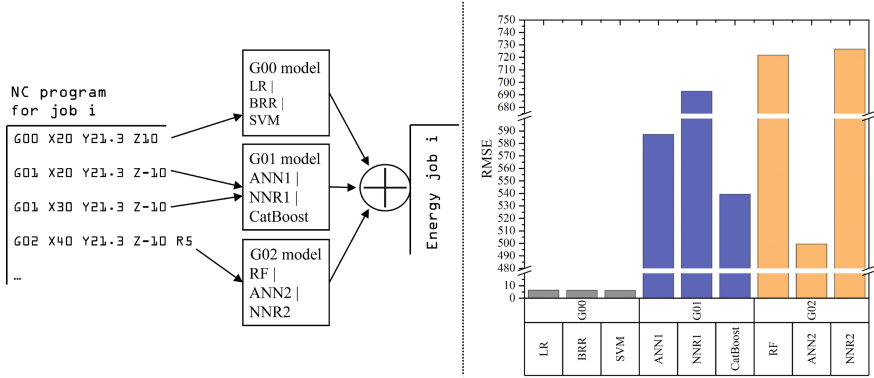
|                          | G00  |      | G01   |       | G02   |       |
|--------------------------|------|------|-------|-------|-------|-------|
|                          | min  | max  | min   | max   | min   | max   |
| <b>X [mm]</b>            | 0    | 678  | 0     | 300   | 0     | 200   |
| <b>Y [mm]</b>            | 0    | 1043 | 0     | 500   | 0     | 300   |
| <b>Z [mm]</b>            | 0    | 30   | 0     | 30    | 0     | 25    |
| <b><i>f</i> [mm/min]</b> | 5100 | 5100 | 200   | 2000  | 200   | 2000  |
| <b><i>S</i> [rpm]</b>    | 0    | 0    | 10000 | 25000 | 10000 | 25000 |

validate the entire approach on the conceptual level in this contribution. Hence, the variations in milling different materials, e.g. aluminum or steel, are left out and the spindle speed is varied to in-/ decrease the energy consumption while executing G01 and G02. As previously mentioned, the Design of Experiment is based on LHS, which provided 200 sample points for each G-code command. These sample points range from 0 to 1 and are multiplied with the maximum of each feature to convert them into values the G-code can work with. A Python script created all necessary G-code lines for all three commands G00, G01, and G02 by using the values LHS provided. During all experiments, the energy of the CNC machine and spindle is measured using a time interval of 0.1 s. The values are then matched to the specific lines of G-code that correspond to the movement or activation of the component. The last step is to train the ML models, which will be explained in Sect. 3.

3 ML-Model Training and Validation

Three separate ML models for G00, G01, and G02 are trained with 80% of the available data. The different model types are trained and evaluated with the metric root mean square error (RMSE) on the remaining 20% of the data. The overall ML approach is shown in Fig. 2 (left). The RMSE of the three best models for each command are shown in Fig. 2 (right). For G00, the three best-performing models regarding the RMSE are Support Vector Regression (SVM-6.25), Bayesian Ridge Regression (BRR-6.38), and Linear Regression (LR-6.4). The best models for predicting the energy consumption of G01 are CatBoost Regression (CatBoost-539.28), Artificial Neural Network (ANN1-587.25), and Nearest Neighbor Regression (NNR1-692.83). For G02, the models with the lowest RMSE are again Artificial Neural Network (ANN2-499.44), Random Forest Regression (RF-721.9), and Nearest Neighbor Regression (NNR2-726.71).

The RMSE of G01 and G02 is significantly higher than the RMSE of G00 as the spindle speed has a high impact on the energy consumption, and for rapid movement commands (G00) the spindle is normally off. After combining the three best models to get the ensemble model, the total RMSE is 478.08.



**Fig. 2.** G-code evaluation principle using an ensemble model approach (left); performance comparison of different ML models by RMSE measure

## 4 Scheduling Results and Discussion

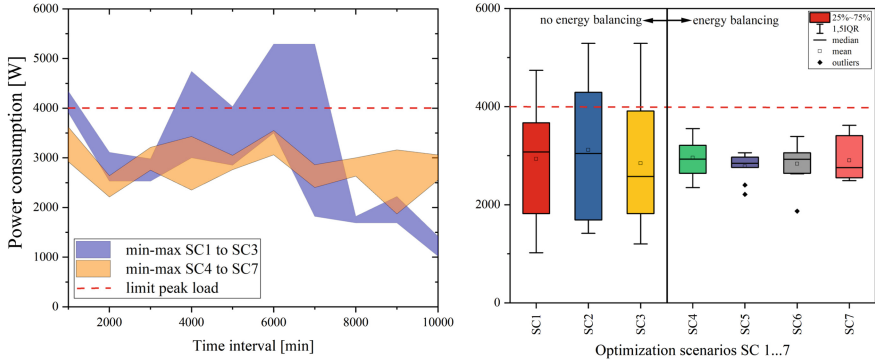
The ML ensemble model results in an additional data point regarding the specific energy consumption of each machining job. Based on this, we developed a JSS algorithm with two objectives (see Eqs. 1 and 2) with  $J_i$  being the processing times of each production order,  $n$  the total number of orders,  $E$  the total energy consumption, and  $\mu$  the mean value of the total energy consumption:

$$\min_{\text{Makespan}} = \sum_{i=1}^n J_i \quad (1)$$

$$\min_{\text{Var}(E)} = \frac{\sqrt{\text{Var}(E)}}{\mu} \quad (2)$$

The JSS algorithm is based on our previous research presenting a cascaded JSS approach for highly autonomous production systems [12] utilizing CP-SAT (constraint programming and Boolean satisfiability), which has a high computational effort. The current JSS algorithm further develops the approach from [12] by a new metaheuristic called Micro Evolutionary Particle Swarm Optimization (MEPSO) [13]. MEPSO is a modified Particle Swarm Optimization algorithm that includes Micropopulations, which lead to a decrease in computational complexity, an accelerated convergence rate, simplified implementation, and prevention of premature convergence. To verify the proposed JSS algorithm, we created seven scenarios SC1 to SC7, each with 16 production orders containing data on processing times, order quantity, due dates, and energy consumption. SC1, SC2, and SC3 do not consider energy consumption and are only optimized by makespan. To flatten out the energy demand, a time interval of 1000 min was chosen, which roughly translates into two work shifts in a manufacturing company. For SC4 to SC7, the JSS algorithm considers energy output for all machines and minimizes the variance between all intervals. The results show peak shaving

for SC4 to SC7 under an exemplary limiting peak load of 4 kW, whereas SC1, SC2, and SC3 form a peak in different time intervals (see Fig. 3, left). The algorithm flattens the energy curves of scenarios 4 to 7 to reach a standard deviation of less than 15% from the mean energy consumption of all time intervals. The minimum distance to the limit value is at  $SC4(6000 \text{ minutes}) = 11.25 \%$  (see Fig. 3, right).



**Fig. 3.** Results of the energy consumption over all time intervals and scenarios SC1 to SC7 (left); box-plot of the assigned jobs and their energy consumption in each scenario

## 5 Conclusions, Limitations and Further Research

The presented contribution faces an innovative mix of methods regarding energy balancing or peak shaving in CNC machining for autonomous machine fleets by combining G-code analysis with ML and JSS. At this moment, an ensemble ML model is trained and validated with a minimum of experimental effort to predict the energy demand of an NC program. As a result, a customer order/machining job contains the energy consumption as additional information, which is then used to perform a JSS algorithm. The objective of this algorithm is to smooth the energy consumption of a multitude of jobs to avoid electricity load peaks over a specified production period. This will help to adapt to varying energy prices set by the power supplier to increase energy output when prices are lower and decrease power output when prices are higher. The approach presented is currently limited to the three base G-code operations G00, G01, and G02 as well as spindle on/off and motion parameters. As of now, other parameters or external boundary conditions such as material to be machined or different M-commands e.g. the cooling pump on/off are not included in the model. This is not a problem, since the chosen ML models are capable of training and validating energy consumption even without actual machining. This makes a good baseline to further enhance the ensemble ML model by steadily adding more features generated by machining actual parts. Also worth noting is the expansion of

different machining equipment like robots e.g. which also contribute to power consumption. These resources can not only be optimized by power but also integrated into the JSS to further enhance other variables like makespan or cycle time.

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## References

1. Negi PK, Ram M, Yadav OP (2018) 1 CNC machine and its importance. In: Basics of CNC Programming. River Publishers, pp 1–16
2. Anderberg S, Kara S (2009) Energy and cost efficiency in CNC machining. In: The 7th CIRP Conference on Sustainable Manufacturing
3. Gong X, De Pessemier T, Martens L, Joseph W (2019) Energy-and labor-aware flexible job shop scheduling under dynamic electricity pricing: a many-objective optimization investigation. *J Clean Prod* 209:1078–1094
4. Borgia S, Pellegrinelli S, Bianchi G, Leonesio M (2014) A reduced model for energy consumption analysis in milling. *Proced CIRP* 17:529–534. Variety Management in Manufacturing
5. Bhinge R, Park J, Law KH, Dornfeld DA, Helu M, Rachuri S (2016) Toward a generalized energy prediction model for machine tools. *J Manufact Sci Eng* 139(4)
6. Shin SJ, Woo J, Rachuri S (2017) Energy efficiency of milling machining: component modeling and online optimization of cutting parameters. *J Clean Prod* 161:12–29
7. Lv J, Tang R, Jia S, Liu Y (2016) Experimental study on energy consumption of computer numerical control machine tools. *J Clean Prod* 112:3864–3874
8. Edem IF, Mativenga PT (2017) Modelling of energy demand from computer numerical control (CNC) toolpaths. *J Clean Prod* 157:310–321
9. He Y, Liu F, Wu T, Zhong F, Peng B (2012) Analysis and estimation of energy consumption for numerical control machining. *Proceed Institut Mech Eng, Part B: J Eng Manuf* 226(2):255–266
10. Morris MD, Mitchell TJ (1995) Exploratory designs for computational experiments. *J Statis Plann Infer* 43(3):381–402
11. Pholdee N, Bureerat S (2015) An efficient optimum Latin hypercube sampling technique based on sequencing optimisation using simulated annealing. *Int J Syst Sci* 46(10):1780–1789
12. Miller E, Kaupp T, Schmitt J (2022) Cascaded scheduling for highly autonomous production cells with agvs. In: Global Conference on Sustainable Manufacturing, pp 383–390. Springer
13. Solano-Rojas BJ, Villalón-Fonseca R, Batres R (2023) Micro evolutionary particle swarm optimization (MEPSO): a new modified metaheuristic. *Syst Soft Comp*, 200057



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